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Organic light emitting diode display device including multiple emitting material layer and method of fabricating the same

Abstract

An organic light emitting diode display device includes a substrate having a plurality of subpixels; and an emitting material layer in the plurality of subpixels on the substrate, wherein the emitting material layer in at least one of the plurality of subpixels comprises: a first emitting material layer including a first host and a dopant; a second emitting material layer on the first emitting material layer and including a second host and the dopant; and a third emitting material layer on the second emitting material layer and including one of the dopant and the second host.

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USPC: None

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims the priority of Korean Patent Application No. 10-2021-0189866 filed on Dec. 28, 2021, which is hereby incorporated by reference in its entirety.

BACKGROUND

Field of the Disclosure

(2) The present disclosure relates to an organic light emitting diode display device, and more particularly, to an organic light emitting diode display device and a method of fabricating the same where an efficiency is improved by sequentially forming first to third emitting material layers of a host and a dopant using a single chamber.

Description of the Background

(3) Recently, with the advent of an information-oriented society and as the interest in information displays for processing and displaying a massive amount of information and the demand for portable information media have increased, a display field has rapidly advanced. Thus, various light and thin flat panel display devices have been developed and highlighted.

(4) Among the various flat panel display devices, an organic light emitting diode (OLED) display device is an emissive type device and does not include a backlight unit used in a non-emissive type device such as a liquid crystal display (LCD) device. As a result, the OLED display device has advantages in a viewing angle, a contrast ratio and a power consumption to be applied to various fields.

(5) In the OLED display device, an efficiency is improved by forming an emitting material layer as a multiple layer using two hosts. Since the two hosts use two independent chambers, a fabrication process becomes complicated. Further, deterioration is caused by a contamination during a transfer or a misalignment of a fine metal mask.

SUMMARY

(6) Accordingly, the present disclosure is directed to an organic light emitting diode display device and a method of fabricating the same that substantially obviate one or more of the problems due to

limitations and disadvantages of the related art.

(7) More specifically, the present disclosure is to provide an organic light emitting display device and a method of fabricating the organic light emitting diode display device where an efficiency is improved and a productivity increases by forming a triple layered emitting material layers of a first emitting material layer of a first host and a dopant, a second emitting material layer of a second host and a dopant and a third emitting material layer of a dopant in a single chamber.

(8) The present disclosure is also to provide an organic light emitting diode display device and a method of fabricating the organic light emitting diode display device where an efficiency is improved and a productivity increases by forming a triple layered emitting material layers of a first emitting material layer of a first host and a dopant, a second emitting material layer of a second host and a dopant and a third emitting material layer of a second host in a single chamber.

(9) Additional features and advantages of the disclosure will be set forth in the description which follows, and in part will be apparent from the description, or may be learned by practice of the disclosure. These and other advantages of the disclosure will be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.

(10) To achieve these and other advantages and in accordance with the purpose of the present disclosure, as embodied and broadly described herein, an organic light emitting diode display device includes a substrate having a plurality of subpixels; and an emitting material layer in the plurality of subpixels on the substrate, wherein the emitting material layer in at least one of the plurality of subpixels comprises: a first emitting material layer including a first host and a dopant; a second emitting material layer on the first emitting material layer and including a second host and the dopant; and a third emitting material layer on the second emitting material layer and including one of the dopant and the second host.

(11) In another aspect of the present disclosure, a method of fabricating an organic light emitting diode display device includes disposing a substrate in a chamber including a first evaporation source transmitting a first host to a first region, a second evaporation source transmitting a second host to a second region separated from the first region, a third evaporation source transmitting a dopant to a third region corresponding to a sum of the first and second regions and first to third shutters corresponding to the first to third evaporation sources, respectively, the substrate disposed in the first region: forming a first emitting material layer by depositing the first host and the dopant of the first and third evaporation sources on the substrate while the first and third shutters have an open state; transferring the substrate to the second region and forming a second emitting material layer by depositing the second host and the dopant of the second and third evaporation sources on the substrate while the second and third shutters have the open state; and forming a third emitting material layer by depositing the dopant of the third evaporation source on the substrate while the second shutter has a closed state and the third shutter has the open state.

(12) In a further aspect of the present disclosure, a method of fabricating an organic light emitting diode display device includes disposing a substrate in a chamber including a first evaporation source transmitting a first host to a first region, a second evaporation source transmitting a second host to a second region separated from the first region, a third evaporation source transmitting a dopant to a third region corresponding to a sum of the first and second regions and first to third shutters corresponding to the first to third evaporation sources, respectively, the substrate disposed in the first region: forming a first emitting material layer by depositing the first host and the dopant of the first and third evaporation sources on the substrate while the first and third shutters have an open state; transferring the substrate to the second region and forming a second emitting material layer by depositing the second host and the dopant of the second and third evaporation sources on the substrate while the second and third shutters have the open state; and forming a third emitting material layer by depositing the second host of the second evaporation source on the substrate while the second shutter has the open state and the third shutter has a closed state.

(13) It is to be understood that both the foregoing general description and the following detailed description are explanatory and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification, illustrate aspects of the disclosure and together with the description serve to explain the principles of the disclosure.

(2) In the drawings:

(3) FIG. **1** is a cross-sectional view showing a multiple layered emitting material layer of an organic light emitting diode display device according to a first aspect of the present disclosure;

(4) FIG. **2** is a view illustrating a method of fabricating a multiple layered emitting material layer of an organic light emitting diode display device according to a first aspect of the present disclosure;

(5) FIG. **3** is an energy band diagram showing a multiple layered emitting material layer of an organic light emitting diode display device according to a first aspect of the present disclosure;

(6) FIG. **4** is a cross-sectional view showing a multiple layered emitting material layer of an organic light emitting diode display device according to a second aspect of the present disclosure;

(7) FIG. **5** is a view illustrating a method of fabricating a multiple layered emitting material layer of an organic light emitting diode display device according to a second aspect of the present disclosure;

(8) FIG. **6** is an energy band diagram showing a multiple layered emitting material layer of an organic light emitting diode display device according to a second aspect of the present disclosure;

(9) FIG. **7** is a view showing an organic light emitting diode display device according to a third aspect of the present disclosure;

(10) FIG. **8** is a view showing an organic light emitting diode display device according to a fourth aspect of the present disclosure;

(11) FIG. **9** is a view showing an organic light emitting diode display device according to a fifth aspect of the present disclosure;

(12) FIG. **10** is a view showing an organic light emitting diode display device according to a sixth aspect of the present disclosure;

(13) FIG. **11** is a view showing an organic light emitting diode display device according to a seventh aspect of the present disclosure;

(14) FIG. **12** is a view showing an organic light emitting diode display device according to an eighth aspect of the present disclosure; and

(15) FIG. **13** is a view showing an organic light emitting diode display device according to a ninth aspect of the present disclosure.

DETAILED DESCRIPTION

(16) Advantages and features of the present disclosure, and implementation methods thereof will be clarified through following example aspects described with reference to the accompanying drawings. The present disclosure may, however, be embodied in different forms and should not be construed as limited to the example aspects set forth herein. Rather, these example aspects are provided so that this disclosure may be sufficiently thorough and complete to assist those skilled in the art to fully understand the scope of the present disclosure. Further, the present disclosure is only defined by scopes of claims.

(17) A shape, a size, a ratio, an angle, and a number disclosed in the drawings for describing aspects of the present disclosure are merely an example. Thus, the present disclosure is not limited to the illustrated details. Like reference numerals refer to like elements throughout. In the following

description, when the detailed description of the relevant known function or configuration is determined to unnecessarily obscure an important point of the present disclosure, the detailed description of such known function or configuration may be omitted. In a case where terms "comprise," "have," and "include" described in the present specification are used, another part may be added unless a more limiting term, such as "only," is used. The terms of a singular form may include plural forms unless referred to the contrary.
(18) In construing an element, the element is construed as including an error or tolerance range even where no explicit description of such an error or tolerance range.
(19) In describing a position relationship, when a position relation between two parts is described as, for example, "on," "over," "under," or "next," one or more other parts may be disposed between the two parts unless a more limiting term, such as "just" or "direct(ly)," is used.
(20) It will be understood that, although the terms "first," "second," etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure.
(21) Features of various aspects of the present disclosure may be partially or overall coupled to or combined with each other, and may be variously inter-operated with each other and driven technically as those skilled in the art can sufficiently understand. Aspects of the present disclosure may be carried out independently from each other, or may be carried out together in co-dependent relationship.
(22) Hereinafter, an organic light emitting diode display device and a method of fabricating the same according to aspects of the present disclosure will be described in detail with reference to the accompanying drawings. In the following description, like reference numerals designate like elements throughout. When a detailed description of well-known functions or configurations related to this document is determined to unnecessarily cloud a gist of the inventive concept, the detailed description thereof will be omitted or will be made brief.
(23) FIG. **1** is a cross-sectional view showing a multiple layered emitting material layer of an organic light emitting diode display device according to a first aspect of the present disclosure, FIG. **2** is a view illustrating a method of fabricating a multiple layered emitting material layer of an organic light emitting diode display device according to a first aspect of the present disclosure, and FIG. **3** is an energy band diagram showing a multiple layered emitting material layer of an organic light emitting diode display device according to a first aspect of the present disclosure.
(24) In FIG. **1**, a multiple layered emitting material layer (EML) of an organic light emitting diode (OLED) display device according to a first aspect of the present disclosure includes first, second and third emitting material layers (EMLs) **130**, **132** and **134**. The first EML **130** includes a first host H**1** and a dopant D, the second EML **132** includes a second host H**2** and the dopant D, and the third EML **134** includes the dopant D.
(25) A thickness of the first EML **130** is greater than or equal to a thickness of the second EML **132**, and a thickness of the third EML **134** is smaller than a thickness of the second EML **132**.
(26) For example, the multiple layered EML of the first, second and third EMLs **130**, **132** and **134** may be disposed in one of red, green and blue subpixels, and the first, second and third EMLs **130**, **132** and **134** may be formed using a single chamber of an evaporation apparatus.
(27) In FIG. **2**, a single chamber CH of an evaporation apparatus includes first, second and third evaporation sources SS**1**, SS**2** and SS**3** and first, second and third shutters ST**1**, ST**2** and ST**3** corresponding to the first, second and third evaporation sources SS**1**, SS**2** and SS**3**, respectively.
(28) The first evaporation source SS**1** transmits the first host H**1** to a first region A**1**, the second evaporation source SS**2** transmits the second host H**2** to a second region A**2**, and the third evaporation source SS**3** transmits the dopant D to a third region A**3**.
(29) The first and second regions A**1** and A**2** do not overlap each other and are separated from each

other, and the third region A**3** corresponds to a sum of the first and second regions A**1** and A**2**.
(30) In the single chamber CH of the evaporation apparatus, a substrate is disposed in the first region A**1**, and the first host H**1** and the dopant D of the first and third evaporation sources SS**1** and SS**3** are deposited on the substrate while the first and third shutters ST**1** and ST**3** have an open state. As a result, the first EML **130** of the first host H**1** and the dopant D is formed in a corresponding subpixel on the substrate.
(31) Next, the substrate is transferred to the second region A**2**, and the second host H**2** and the dopant D of the second and third evaporation sources SS**2** and SS**3** are deposited on the substrate while the second and third shutters ST**2** and ST**3** have an open state. As a result, the second EML **132** of the second host H**2** and the dopant D is formed on the first EML **130**.
(32) Next, the substrate is disposed in the second region A**2**, and the dopant D of the third evaporation sources SS**3** are deposited on the substrate while the second shutter ST**2** is changed to have a closed state and the third shutter ST**3** has the open state. As a result, the third EML **134** of the dopant D is formed on the second EML **132**.
(33) In another aspect, the substrate may be transferred to the first region A**1**, and the dopant D of the third evaporation source SS**3** may be deposited on the substrate while the first shutter ST**1** is changed to have a closed state and the third shutter ST**3** has the open state. As a result, the third EML **134** of the dopant D may be formed on the second EML **132**.
(34) In FIG. **3**, the first, second and third EMLs **130**, **132** and **134** are disposed between an electron blocking layer **126** and a hole blocking layer **140**.
(35) An energy level of a lowest unoccupied molecular orbit (LUMO) of the electron blocking layer **126** is higher than an energy level of an LUMO of the first EML **130**, and an energy level of a highest occupied molecular orbit (HOMO) of the electron blocking layer **126** is lower than an energy level of a HOMO of the first EML **130**.
(36) Energy levels of LUMOs of the first and second EMLs **130** and **132** are the same as each other, and energy levels of HOMOs of the first and second EMLs **130** and **132** are the same as each other.
(37) An energy level of a triplet excited state T**1** of the first host H**1** of the first EML **130** is lower than an energy level of a triplet excited state T**1** of the second host H**2** of the second EML **132**.
(38) Energy levels of triplet excited states T**1** of the dopants of the first and second EMLs **130** and **132** are the same as each other, and energy levels of singlet ground states S**0** of the dopants of the first and second EMLs **130** and **132** are the same as each other.
(39) The first and second hosts H**1** and H**2** of the first and second EMLs **130** and **132** have a bipolar type having a relatively high electron mobility.
(40) The energy level of the LUMO of the third EML **134** is lower than the energy level of the LUMO of the second EML **132** and is higher than the energy level of the LUMO of the hole blocking layer **140**. The energy level of the HOMO of the third EML **134** is lower than the energy level of the HOMO of the second EML **132** and is higher than the energy level of the HOMO of the hole blocking layer **140**.
(41) The energy level of the LUMO of the third EML **134** has a value between the energy level of the LUMO of the second EML **132** and the energy level of the LUMO of the hole blocking layer **140**. The energy level of the HOMO of the third EML **134** has a value between the energy level of the HOMO of the second EML **132** and the energy level of the HOMO of the hole blocking layer **140**.
(42) In the multiple layered EML, a hole h of the hole blocking layer **140** is transmitted to the first EML **130** through the third EML **134** and the second EML **132**, and an electron e of the electron blocking layer **126** is transmitted to the first EML **130**. As a result, a recombination zone of the hole h and the electron e is formed in the first EML **130** to emit a light.
(43) Since the energy level of the LUMO of the dopant D of the third EML **134** has a value between the energy level of the LUMO of the second host H**2** and the dopant D of the second EML

132 and the energy level of the LUMO of the hole blocking layer **140**, an electron mobility increases and the electron is transmitted to the first EML **130** promptly to improve an emission efficiency.
(44) Further, since the multiple layered EML is formed in the single chamber, the fabrication process is simplified and deterioration due to contamination during transferring or misalignment of a fine metal mask is prevented.
(45) Although the first, second and third EMLs **130**, **132** and **134** exemplarily include the same kind of dopant D in the first aspect, the first, second and third EMLs **130**, **132** and **134** may include a dopant different from each other in another aspect.
(46) The third EML may include the second host in another aspect, and it will be illustrated with reference to drawings.
(47) FIG. **4** is a cross-sectional view showing a multiple layered emitting material layer of an organic light emitting diode display device according to a second aspect of the present disclosure, FIG. **5** is a view illustrating a method of fabricating a multiple layered emitting material layer of an organic light emitting diode display device according to a second aspect of the present disclosure, and FIG. **6** is an energy band diagram showing a multiple layered emitting material layer of an organic light emitting diode display device according to a second aspect of the present disclosure.
(48) In FIG. **4**, a multiple layered emitting material layer (EML) of an organic light emitting diode (OLED) display device according to a second aspect of the present disclosure includes first, second and third emitting material layers (EMLs) **230**, **232** and **236**.
(49) A thickness of the first EML **230** is greater than or equal to a thickness of the second EML **232**, and a thickness of the third EML **236** is smaller than a thickness of the second EML **232**.
(50) For example, the multiple layered EML of the first, second and third EMLs **230**, **232** and **236** may be disposed in one of red, green and blue subpixels, and the first, second and third EMLs **230**, **232** and **236** may be formed using a single chamber of an evaporation apparatus.
(51) In FIG. **5**, a single chamber CH of an evaporation apparatus includes first, second and third evaporation sources SS**1**, SS**2** and SS**3** and first, second and third shutters ST**1**, ST**2** and ST**3** corresponding to the first, second and third evaporation sources SS**1**, SS**2** and SS**3**, respectively.
(52) The first evaporation source SS**1** transmits the first host H**1** to a first region A**1**, the second evaporation source SS**2** transmits the second host H**2** to a second region A**2**, and the third evaporation source SS**3** transmits the dopant D to a third region A**3**.
(53) The first and second regions A**1** and A**2** do not overlap each other, and the third region A**3** corresponds to a sum of the first and second regions A**1** and A**2**.
(54) In the single chamber CH of the evaporation apparatus, a substrate is disposed in the first region A**1**, and the first host H**1** and the dopant D of the first and third evaporation sources SS**1** and SS**3** are deposited on the substrate while the first and third shutters ST**1** and ST**3** have an open state. As a result, the first EML **230** of the first host H**1** and the dopant D is formed in a corresponding subpixel on the substrate.
(55) Next, the substrate is transferred to the second region A**2**, and the second host H**2** and the dopant D of the second and third evaporation sources SS**2** and SS**3** are deposited on the substrate while the second and third shutters ST**2** and ST**3** have an open state. As a result, the second EML **232** of the second host H**2** and the dopant D is formed on the first EML **230**.
(56) Next, the substrate is disposed in the second region A**2**, and the second host H**2** of the second evaporation sources SS**2** is deposited on the substrate while the third shutter ST**3** is changed to have a closed state and the second shutter ST**2** has an open state. As a result, the third EML **236** of the second host H**2** is formed on the second EML **232**.
(57) In FIG. **6**, the first, second and third EMLs **230**, **232** and **236** are disposed between an electron blocking layer **226** and a hole blocking layer **240**.
(58) An energy level of a lowest unoccupied molecular orbit (LUMO) of the electron blocking layer **226** is higher than an energy level of an LUMO of the first EML **230**, and an energy level of a

highest occupied molecular orbit (HOMO) of the electron blocking layer **226** is lower than an energy level of a HOMO of the first EML **230**.
(59) Energy levels of LUMOs of the first, second and third EMLs **230**, **232** and **236** are the same as each other, and energy levels of HOMOs of the first, second and third EMLs **230**, **232** and **236** are the same as each other.
(60) An energy level of a triplet excited state T**1** of the first host H**1** of the first EML **230** is lower than an energy level of a triplet excited state T**1** of the second host H**2** of the second EML **232**.
(61) Energy levels of triplet excited states T**1** of the dopants of the first and second EMLs **230** and **232** are the same as each other, and energy levels of singlet ground states S**0** of the dopants of the first and second EMLs **230** and **232** are the same as each other.
(62) The energy level of the LUMO of the third EML **236** is higher than the energy level of the LUMO of the hole blocking layer **240**, and the energy level of the HOMO of the third EML **236** is higher than the energy level of the HOMO of the hole blocking layer **240**.
(63) The second host H**2** of the second and third EMLs **232** and **236** have a bipolar type having a relatively high electron mobility.
(64) In the multiple layered EML, a hole h of the hole blocking layer **240** is transmitted to the first EML **230** through the third EML **236** and the second EML **232**, and an electron e of the electron blocking layer **226** is transmitted to the first EML **230**. As a result, a recombination zone of the hole h and the electron e is formed in the first EML **230** to emit a light.
(65) Since the second host H**2** of the third emitting material layer **236** is formed of a bipolar type material capable of transmitting an exciton with a relatively high electron mobility, the electron mobility is improved and the electron e is promptly transmitted to the first EML **230**. As a result, an emission efficiency is improved.
(66) Further, since the multiple layered EML is formed in the single chamber, the fabrication process is simplified and deterioration due to contamination during transferring or misalignment of a fine metal mask is prevented.
(67) Although the first and second EMLs **230** and **232** exemplarily include the same kind of dopant D in the second aspect, the first and second EMLs **230** and **232** may include a dopant different from each other in another aspect.
(68) The multiple layered EML may be disposed in at least one of the red, green and blue subpixels in another aspect, and it will be illustrated with reference to drawings.
(69) FIG. **7** is a view showing an organic light emitting diode display device according to a third aspect of the present disclosure. A part of the third aspect the same as the first and second aspects will be omitted.
(70) In FIG. **7**, an organic light emitting diode (OLED) display device **310** according to a third aspect of the present disclosure includes red, green and blue subpixels SPr, SPg and SPb.
(71) An anode (not shown) is disposed in each of the red, green and blue subpixels SPr, SPg and SPb on a substrate (not shown). A hole injecting layer (HIL) **320** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the anode, and a hole transporting layer (HTL) **322** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the HIL **320**.
(72) In another aspect, a positive type hole transporting layer (P-HTL) instead of the HIL **320** may be disposed between the anode and the HTL **322**.
(73) A red hole transporting layer (HTL) **324***r* is selectively disposed in the red subpixel SPr on the HTL **322**, and a green hole transporting layer (HTL) **324***g* is selectively disposed in the green subpixel SPg on the HTL **322**.
(74) The red HTL **324***r* and the green HTL **324***g* may adjust a length of an optical path in the red, green and blue subpixels SPr, SPg and SPb to obtain a micro cavity.
(75) An electron blocking layer (EBL) **326** is disposed on a whole of the red HTL **324***r* of the red subpixel SPr, the green HTL **324***g* of the green subpixel SPg and the HTL **322** of the blue subpixel SPb.

(76) A red emitting material layer (EML) **330***r* is selectively disposed on the EBL **326** of the red subpixel SPr, and a green emitting material layer (EML) **330***g* is disposed on the EBL **326** of the green subpixel SPg.
(77) A first blue emitting material layer (EML) **330***b*, a second blue emitting material layer (EML) **332***b* and a third blue emitting material layer (EML) **334***b* may be sequentially disposed on the EBL **326** of the blue subpixel SPb, or a first blue emitting material layer (EML) **330***b*, a second blue emitting material layer (EML) **332***b* and a third blue emitting material layer (EML) **336***b* may be sequentially disposed on the EBL **326** of the blue subpixel SPb.
(78) A thickness of the first blue EML **330***b* is greater than or equal to a thickness of the second blue EML **332***b*, and a thickness of the third blue EML **334***b* and **336***b* is smaller than a thickness of the second blue EML **332***b*.
(79) A thickness of the green EML **330***g* is greater than a total thickness of the first blue EML **330***b*, the second blue EML **332***b* and the third blue EML **334***b* or a total thickness of the first blue EML **330***b*, the second blue EML **332***b* and the third blue EML **336***b* and is smaller than a thickness of the red EML **330***r*.
(80) The first blue EML **330***b* includes a first blue host and a blue dopant, and the second blue EML **332***b* includes a second blue host and a blue dopant. The third blue EML **334***b* includes a blue dopant, and the third blue EML **336***b* includes a second blue host.
(81) The first blue EML **330***b*, the second blue EML **332***b* and the third blue EML **334***b* may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first blue EML **330***b*, the second blue EML **332***b* and the third blue EML **336***b* may have the same structure as those of the second aspect of FIGS. **4** to **6**.
(82) An energy level of a triplet excited state T**1** of the first blue host of the first blue EML **330***b* is lower than an energy level of a triplet excited state T**1** of the second blue host of the second blue EML **332***b*.
(83) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third blue EML **334***b* has a value between an energy level of a LUMO of the second blue EML **332***b* and an energy level of a LUMO of the HBL **340**, and an energy level of a highest occupied molecular orbit (HOMO) of the third blue EML **334***b* has a value between an energy level of a HOMO of the second blue EML **332***b* and an energy level of a HOMO of the HBL **340**.
(84) As a result, an electron is promptly transmitted to the first blue EML **330***b* to improve an emission efficiency.
(85) The second blue host of the third blue EML **336***b* may include a bipolar type material.
(86) As a result, an electron is promptly transmitted to the first blue EML **330***b* to improve an emission efficiency.
(87) Further, since the multiple layered EML is formed in the single chamber, the fabrication process is simplified and deterioration due to contamination during transferring or misalignment of a fine metal mask is prevented.
(88) A hole blocking layer (HBL) **340** is disposed on a whole of the red EML **330***r* of the red subpixel SPr, the green EML **330***g* of the green subpixel SPg and the third blue EML **334***b* and **336***b* of the blue subpixel SPb, and an electron transporting layer (ETL) **342** is disposed on the HBL **340** in the red, green and blue subpixels SPr, SPg and SPb.
(89) First and second cathodes **344** and **346** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the ETL **342**, and first and second capping layers **350** and **352** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the second cathode **346**.
(90) FIG. **8** is a view showing an organic light emitting diode display device according to a fourth aspect of the present disclosure. A part of the fourth aspect the same as the first and second aspects will be omitted.
(91) In FIG. **8**, an organic light emitting diode (OLED) display device **410** according to a fourth

aspect of the present disclosure includes red, green and blue subpixels SPr, SPg and SPb.

(92) An anode (not shown) is disposed in each of the red, green and blue subpixels SPr, SPg and SPb on a substrate (not shown). A hole injecting layer (HIL) **420** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the anode, and a hole transporting layer (HTL) **422** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the HIL **420**.

(93) In another aspect, a positive type hole transporting layer (P-HTL) instead of the HIL **420** may be disposed between the anode and the HTL **422**.

(94) A red hole transporting layer (HTL) **424***r* is selectively disposed in the red subpixel SPr on the HTL **422**, and a green hole transporting layer (HTL) **424***g* is selectively disposed in the green subpixel SPg on the HTL **422**.

(95) The red HTL **424***r* and the green HTL **424***g* may adjust a length of an optical path in the red, green and blue subpixels SPr, SPg and SPb to obtain a micro cavity.

(96) An electron blocking layer (EBL) **426** is disposed on a whole of the red HTL **424***r* of the red subpixel SPr, the green HTL **424***g* of the green subpixel SPg and the HTL **422** of the blue subpixel SPb.

(97) A red emitting material layer (EML) **430***r* is selectively disposed on the EBL **426** of the red subpixel SPr, and a blue emitting material layer (EML) **430***b* is disposed on the EBL **426** of the blue subpixel SPb.

(98) A first green emitting material layer (EML) **430***g*, a second green emitting material layer (EML) **432***g* and a third green emitting material layer (EML) **434***g* may be sequentially disposed on the EBL **426** of the green subpixel SPg, or a first green emitting material layer (EML) **430***g*, a second green emitting material layer (EML) **432***g* and a third green emitting material layer (EML) **436***g* may be sequentially disposed on the EBL **426** of the green subpixel SPg.

(99) A thickness of the first green EML **430***g* is greater than or equal to a thickness of the second green EML **432***g*, and a thickness of the third green EML **434***g* and **436***g* is smaller than a thickness of the second green EML **432***g*.

(100) A total thickness of the first green EML **430***g*, the second green EML **432***g* and the third green EML **434***g* or a total thickness of the first green EML **430***g*, the second green EML **432***g* and the third green EML **436***g* is greater than a thickness of the blue EML **430***b* and is smaller than a thickness of the red EML **430***r*.

(101) The first green EML **430***g* includes a first green host and a green dopant, and the second green EML **432***g* includes a second green host and a green dopant. The third green EML **434***g* includes a green dopant, and the third green EML **436***g* includes a second green host.

(102) The first green EML **430***g*, the second green EML **432***g* and the third green EML **434***g* may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first green EML **430***g*, the second green EML **432***g* and the third green EML **436***g* may have the same structure as those of the second aspect of FIGS. **4** to **6**.

(103) An energy level of a triplet excited state T**1** of the first green host of the first green EML **430***g* is lower than an energy level of a triplet excited state T**1** of the second green host of the second green EML **432***g*.

(104) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third green EML **434***g* has a value between an energy level of a LUMO of the second green EML **432***g* and an energy level of a LUMO of the HBL **440**, and an energy level of a highest occupied molecular orbit (HOMO) of the third green EML **434***g* has a value between an energy level of a HOMO of the second green EML **432***g* and an energy level of a HOMO of the HBL **440**.

(105) As a result, an electron is promptly transmitted to the first green EML **430***g* to improve an emission efficiency.

(106) The second green host of the third green EML **436***g* may include a bipolar type material.

(107) As a result, an electron is promptly transmitted to the first green EML **430***g* to improve an emission efficiency.

(108) Further, since the multiple layered EML is formed in the single chamber, the fabrication process is simplified and deterioration due to contamination during transferring or misalignment of a fine metal mask is prevented.
(109) A hole blocking layer (HBL) **440** is disposed on a whole of the red EML **430***r* of the red subpixel SPr, the third green EML **434***g* and **436***g* of the green subpixel SPg and the blue EML **430***b* of the blue subpixel SPb, and an electron transporting layer (ETL) **442** is disposed on the HBL **440** in the red, green and blue subpixels SPr, SPg and SPb.
(110) First and second cathodes **444** and **446** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the ETL **442**, and first and second capping layers **450** and **452** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the second cathode **446**.
(111) FIG. **9** is a view showing an organic light emitting diode display device according to a fifth aspect of the present disclosure. A part of the fifth aspect the same as the first and second aspects will be omitted.
(112) In FIG. **9**, an organic light emitting diode (OLED) display device **510** according to a fifth aspect of the present disclosure includes red, green and blue subpixels SPr, SPg and SPb.
(113) An anode (not shown) is disposed in each of the red, green and blue subpixels SPr, SPg and SPb on a substrate (not shown). A hole injecting layer (HIL) **520** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the anode, and a hole transporting layer (HTL) **522** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the HIL **520**.
(114) In another aspect, a positive type hole transporting layer (P-HTL) instead of the HIL **520** may be disposed between the anode and the HTL **522**.
(115) A red hole transporting layer (HTL) **524***r* is selectively disposed in the red subpixel SPr on the HTL **522**, and a green hole transporting layer (HTL) **524***g* is selectively disposed in the green subpixel SPg on the HTL **522**.
(116) The red HTL **524***r* and the green HTL **524***g* may adjust a length of an optical path in the red, green and blue subpixels SPr, SPg and SPb to obtain a micro cavity.
(117) An electron blocking layer (EBL) **526** is disposed on a whole of the red HTL **524***r* of the red subpixel SPr, the green HTL **524***g* of the green subpixel SPg and the HTL **522** of the blue subpixel SPb.
(118) A first red emitting material layer (EML) **530***r*, a second red emitting material layer (EML) **532***r* and a third red emitting material layer (EML) **534***r* may be sequentially disposed on the EBL **526** of the red subpixel SPr, or a first red emitting material layer (EML) **530***r*, a second red emitting material layer (EML) **532***r* and a third red emitting material layer (EML) **536***r* may be sequentially disposed on the EBL **526** of the red subpixel SPr.
(119) A thickness of the first red EML **530***r* is greater than or equal to a thickness of the second red EML **532***r*, and a thickness of the third red EML **534***r* and **536***r* is smaller than a thickness of the second red EML **532***r.*
(120) The green EML **530***g* is selectively disposed on the EBL **526** of the green subpixel SPg, and the blue EML **530***b* is selectively disposed on the EBL **526** of the blue subpixel SPb.
(121) A thickness of the green EML **530***g* is greater than a thickness of the blue EML **530***b* and is smaller than a total thickness of the first red EML **530***r*, the second red EML **532***r* and the third red EML **534***r* or a total thickness of the first red EML **530***r*, the second red EML **532***r* and the third red EML **536***r.*
(122) The first red EML **530***r* includes a first red host and a red dopant, and the second red EML **532***r* includes a second red host and a red dopant. The third red EML **534***r* includes a red dopant, and the third red EML **536***r* includes a second red host.
(123) The first red EML **530***r*, the second red EML **532***r* and the third red EML **534***r* may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first red EML **530***r*, the second red EML **532***r* and the third red EML **536***r* may have the same structure as those of the second

aspect of FIGS. **4** to **6**.
(124) An energy level of a triplet excited state T**1** of the first red host of the first red EML **530***r* is lower than an energy level of a triplet excited state T**1** of the second red host of the second red EML **532***r*.
(125) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third red EML **534***r* has a value between an energy level of a LUMO of the second red EML **532***r* and an energy level of a LUMO of the HBL **540**, and an energy level of a highest occupied molecular orbit (HOMO) of the third red EML **534***r* has a value between an energy level of a HOMO of the second red EML **532***r* and an energy level of a HOMO of the HBL **540**.
(126) As a result, an electron is promptly transmitted to the first red EML **530***r* to improve an emission efficiency.
(127) The second red host of the third red EML **536***r* may include a bipolar type material.
(128) As a result, an electron is promptly transmitted to the first red EML **530***r* to improve an emission efficiency.
(129) Further, since the multiple layered EML is formed in the single chamber, the fabrication process is simplified and deterioration due to contamination during transferring or misalignment of a fine metal mask is prevented.
(130) A hole blocking layer (HBL) **540** is disposed on a whole of the third red EML **534***r* and **536***r* of the red subpixel SPr, the green EML **530***g* of the green subpixel SPg and the blue EML **530***b* of the blue subpixel SPb, and an electron transporting layer (ETL) **542** is disposed on the HBL **540** in the red, green and blue subpixels SPr, SPg and SPb.
(131) First and second cathodes **544** and **546** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the ETL **542**, and first and second capping layers **550** and **552** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the second cathode **546**.
(132) FIG. **10** is a view showing an organic light emitting diode display device according to a sixth aspect of the present disclosure. A part of the sixth aspect the same as the first and second aspects will be omitted.
(133) In FIG. **10**, an organic light emitting diode (OLED) display device **610** according to a sixth aspect of the present disclosure includes red, green and blue subpixels SPr, SPg and SPb.
(134) An anode (not shown) is disposed in each of the red, green and blue subpixels SPr, SPg and SPb on a substrate (not shown). A hole injecting layer (HIL) **620** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the anode, and a hole transporting layer (HTL) **622** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the HIL **620**.
(135) In another aspect, a positive type hole transporting layer (P-HTL) instead of the HIL **620** may be disposed between the anode and the HTL **622**.
(136) A red hole transporting layer (HTL) **624***r* is selectively disposed in the red subpixel SPr on the HTL **622**, and a green hole transporting layer (HTL) **624***g* is selectively disposed in the green subpixel SPg on the HTL **622**.
(137) The red HTL **624***r* and the green HTL **624***g* may adjust a length of an optical path in the red, green and blue subpixels SPr, SPg and SPb to obtain a micro cavity.
(138) An electron blocking layer (EBL) **626** is disposed on a whole of the red HTL **624***r* of the red subpixel SPr, the green HTL **624***g* of the green subpixel SPg and the HTL **622** of the blue subpixel SPb.
(139) A red emitting material layer (EML) **630***r* is selectively disposed on the EBL **626** of the red subpixel SPr.
(140) A first green emitting material layer (EML) **630***g*, a second green emitting material layer (EML) **632***g* and a third green emitting material layer (EML) **634***g* may be sequentially disposed on the EBL **626** of the green subpixel SPg, or a first green emitting material layer (EML) **630***g*, a second green emitting material layer (EML) **632***g* and a third green emitting material layer (EML)

636*g* may be sequentially disposed on the EBL **626** of the green subpixel SPg.
(141) A thickness of the first green EML **630***g* is greater than or equal to a thickness of the second green EML **632***g*, and a thickness of the third green EML **634***g* and **636***g* is smaller than a thickness of the second green EML **632***g*.
(142) A first blue emitting material layer (EML) **630***b*, a second blue emitting material layer (EML) **632***b* and a third blue emitting material layer (EML) **634***b* may be sequentially disposed on the EBL **626** of the blue subpixel SPb, or a first blue emitting material layer (EML) **630***b*, a second blue emitting material layer (EML) **632***b* and a third blue emitting material layer (EML) **636***b* may be sequentially disposed on the EBL **626** of the blue subpixel SPb.
(143) A thickness of the first blue EML **630***b* is greater than or equal to a thickness of the second blue EML **632***b*, and a thickness of the third blue EML **634***b* and **636***b* is smaller than a thickness of the second blue EML **632***b*.
(144) A total thickness of the first green EML **630***g*, the second green EML **632***g* and the third green EML **634***g* or a total thickness of the first green EML **630***g*, the second green EML **632***g* and the third green EML **636***g* is greater than a total thickness of the first blue EML **630***b*, the second blue EML **632***b* and the third blue EML **634***b* or a total thickness of the first blue EML **630***b*, the second blue EML **632***b* and the third blue EML **636***b* and is smaller than a thickness of the red EML **630***r*.
(145) The first green EML **630***g* includes a first green host and a green dopant, and the second green EML **632***g* includes a second green host and a green dopant. The third green EML **634***g* includes a green dopant, and the third green EML **636***g* includes a second green host.
(146) The first blue EML **630***b* includes a first blue host and a blue dopant, and the second blue EML **632***b* includes a second blue host and a blue dopant. The third blue EML **634***b* includes a blue dopant, and the third blue EML **636***b* includes a second blue host.
(147) The first green EML **630***g*, the second green EML **632***g* and the third green EML **634***g* may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first green EML **630***g*, the second green EML **632***g* and the third green EML **636***g* may have the same structure as those of the second aspect of FIGS. **4** to **6**.
(148) The first blue EML **630***b*, the second blue EML **632***b* and the third blue EML **634***b* may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first blue EML **630***b*, the second blue EML **632***b* and the third blue EML **636***b* may have the same structure as those of the second aspect of FIGS. **4** to **6**.
(149) An energy level of a triplet excited state T**1** of the first green host of the first green EML **630***g* is lower than an energy level of a triplet excited state T**1** of the second green host of the second green EML **632***g*.
(150) An energy level of a triplet excited state T**1** of the first blue host of the first blue EML **630***b* is lower than an energy level of a triplet excited state T**1** of the second blue host of the second blue EML **632***b*.
(151) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third green EML **634***g* has a value between an energy level of a LUMO of the second green EML **632***g* and an energy level of a LUMO of the HBL **640**, and an energy level of a highest occupied molecular orbit (HOMO) of the third green EML **634***g* has a value between an energy level of a HOMO of the second green EML **632***g* and an energy level of a HOMO of the HBL **640**.
(152) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third blue EML **634***b* has a value between an energy level of a LUMO of the second blue EML **632***b* and an energy level of a LUMO of the HBL **640**, and an energy level of a highest occupied molecular orbit (HOMO) of the third blue EML **634***b* has a value between an energy level of a HOMO of the second blue EML **632***b* and an energy level of a HOMO of the HBL **640**.
(153) As a result, an electron is promptly transmitted to the first green EML **630***g* and the first blue EML **630***b* to improve an emission efficiency.

(154) The second green host of the third green EML **636***g* and the second blue host of the third blue EML **636***b* may include a bipolar type material.
(155) As a result, an electron is promptly transmitted to the first green EML **630***g* and the first blue EML **630***b* to improve an emission efficiency.
(156) Further, since the multiple layered EML is formed in the single chamber, the fabrication process is simplified and deterioration due to contamination during transferring or misalignment of a fine metal mask is prevented.
(157) A hole blocking layer (HBL) **640** is disposed on a whole of the red EML **630***r* of the red subpixel SPr, the third green EML **634***g* and **636***g* of the green subpixel SPg and the third blue EML **634***b* and **636***b* of the blue subpixel SPb, and an electron transporting layer (ETL) **642** is disposed on the HBL **640** in the red, green and blue subpixels SPr, SPg and SPb.
(158) First and second cathodes **644** and **646** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the ETL **642**, and first and second capping layers **650** and **652** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the second cathode **646**.
(159) FIG. **11** is a view showing an organic light emitting diode display device according to a seventh aspect of the present disclosure. A part of the seventh aspect the same as the first and second aspects will be omitted.
(160) In FIG. **11**, an organic light emitting diode (OLED) display device **710** according to a seventh aspect of the present disclosure includes red, green and blue subpixels SPr, SPg and SPb.
(161) An anode (not shown) is disposed in each of the red, green and blue subpixels SPr, SPg and SPb on a substrate (not shown). A hole injecting layer (HIL) **720** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the anode, and a hole transporting layer (HTL) **722** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the HIL **720**.
(162) In another aspect, a positive type hole transporting layer (P-HTL) instead of the HIL **720** may be disposed between the anode and the HTL **722**.
(163) A red hole transporting layer (HTL) **724***r* is selectively disposed in the red subpixel SPr on the HTL **722**, and a green hole transporting layer (HTL) **724***g* is selectively disposed in the green subpixel SPg on the HTL **722**.
(164) The red HTL **724***r* and the green HTL **724***g* may adjust a length of an optical path in the red, green and blue subpixels SPr, SPg and SPb to obtain a micro cavity.
(165) An electron blocking layer (EBL) **726** is disposed on a whole of the red HTL **724***r* of the red subpixel SPr, the green HTL **724***g* of the green subpixel SPg and the HTL **722** of the blue subpixel SPb.
(166) A first red emitting material layer (EML) **730***r*, a second red emitting material layer (EML) **732***r* and a third red emitting material layer (EML) **734***r* may be sequentially disposed on the EBL **726** of the red subpixel SPr, or a first red emitting material layer (EML) **730***r*, a second red emitting material layer (EML) **732***r* and a third red emitting material layer (EML) **736***r* may be sequentially disposed on the EBL **726** of the red subpixel SPr.
(167) A thickness of the first red EML **730***r* is greater than or equal to a thickness of the second red EML **732***r*, and a thickness of the third red EML **734***r* and **736***r* is smaller than a thickness of the second red EML **732***r.*
(168) The green EML **730***g* is selectively disposed on the EBL **726** of the green subpixel SPg.
(169) A first blue emitting material layer (EML) **730***b*, a second blue emitting material layer (EML) **732***b* and a third blue emitting material layer (EML) **734***b* may be sequentially disposed on the EBL **726** of the blue subpixel SPb, or a first blue emitting material layer (EML) **730***b*, a second blue emitting material layer (EML) **732***b* and a third blue emitting material layer (EML) **736***b* may be sequentially disposed on the EBL **726** of the blue subpixel SPb.
(170) A thickness of the first blue EML **730***b* is greater than or equal to a thickness of the second blue EML **732***b*, and a thickness of the third blue EML **734***b* and **736***b* is smaller than a thickness of

the second blue EML **732***b*.
(171) A thickness of the green EML **730***g* is greater than a total thickness of the first blue EML **730***b*, the second blue EML **732***b* and the third blue EML **734***b* or a total thickness of the first blue EML **730***b*, the second blue EML **732***b* and the third blue EML **736***b* and is smaller than a total thickness of the first red EML **730***r*, the second red EML **732***r* and the third red EML **734***r* or a total thickness of the first red EML **730***r*, the second red EML **732***r* and the third red EML **736***r*.
(172) The first red EML **730***r* includes a first red host and a red dopant, and the second red EML **732***r* includes a second red host and a red dopant. The third red EML **734***r* includes a red dopant, and the third red EML **736***r* includes a second red host.
(173) The first blue EML **730***b* includes a first blue host and a blue dopant, and the second blue EML **732***b* includes a second blue host and a blue dopant. The third blue EML **734***b* includes a blue dopant, and the third blue EML **736***b* includes a second blue host.
(174) The first red EML **730***r*, the second red EML **732***r* and the third red EML **734***r* may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first red EML **730***r*, the second red EML **732***r* and the third red EML **736***r* may have the same structure as those of the second aspect of FIGS. **4** to **6**.
(175) The first blue EML **730***b*, the second blue EML **732***b* and the third blue EML **734***b* may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first blue EML **730***b*, the second blue EML **732***b* and the third blue EML **736***b* may have the same structure as those of the second aspect of FIGS. **4** to **6**.
(176) An energy level of a triplet excited state T**1** of the first red host of the first red EML **730***r* is lower than an energy level of a triplet excited state T**1** of the second red host of the second red EML **732***r*.
(177) An energy level of a triplet excited state T**1** of the first blue host of the first blue EML **730***b* is lower than an energy level of a triplet excited state T**1** of the second blue host of the second blue EML **732***b*.
(178) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third red EML **734***r* has a value between an energy level of a LUMO of the second red EML **732***r* and an energy level of a LUMO of the HBL **740**, and an energy level of a highest occupied molecular orbit (HOMO) of the third red EML **734***r* has a value between an energy level of a HOMO of the second red EML **732***r* and an energy level of a HOMO of the HBL **740**.
(179) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third blue EML **734***b* has a value between an energy level of a LUMO of the second blue EML **732***b* and an energy level of a LUMO of the HBL **740**, and an energy level of a highest occupied molecular orbit (HOMO) of the third blue EML **734***b* has a value between an energy level of a HOMO of the second blue EML **732***b* and an energy level of a HOMO of the HBL **740**.
(180) As a result, an electron is promptly transmitted to the first red EML **730***r* and the first blue EML **730***b* to improve an emission efficiency.
(181) The second red host of the third red EML **736***r* and the second blue host of the third blue EML **736***b* may include a bipolar type material.
(182) As a result, an electron is promptly transmitted to the first red EML **730***r* and the first blue EML **730***b* to improve an emission efficiency.
(183) Further, since the multiple layered EML is formed in the single chamber, the fabrication process is simplified and deterioration due to contamination during transferring or misalignment of a fine metal mask is prevented.
(184) A hole blocking layer (HBL) **740** is disposed on a whole of the third red EML **734***r* and **736***r* of the red subpixel SPr, the green EML **730***g* of the green subpixel SPg and the third blue EML **734***b* and **736***b* of the blue subpixel SPb, and an electron transporting layer (ETL) **742** is disposed on the HBL **740** in the red, green and blue subpixels SPr, SPg and SPb.
(185) First and second cathodes **744** and **746** are sequentially disposed in a whole of the red, green

and blue subpixels SPr, SPg and SPb on the ETL **742**, and first and second capping layers **750** and **752** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the second cathode **746**.
(186) FIG. **12** is a view showing an organic light emitting diode display device according to an eighth aspect of the present disclosure. A part of the eighth aspect the same as the first and second aspects will be omitted.
(187) In FIG. **12**, an organic light emitting diode (OLED) display device **810** according to an eighth aspect of the present disclosure includes red, green and blue subpixels SPr, SPg and SPb.
(188) An anode (not shown) is disposed in each of the red, green and blue subpixels SPr, SPg and SPb on a substrate (not shown). A hole injecting layer (HIL) **820** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the anode, and a hole transporting layer (HTL) **822** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the HIL **820**.
(189) In another aspect, a positive type hole transporting layer (P-HTL) instead of the HIL **820** may be disposed between the anode and the HTL **822**.
(190) A red hole transporting layer (HTL) **824***r* is selectively disposed in the red subpixel SPr on the HTL **822**, and a green hole transporting layer (HTL) **824***g* is selectively disposed in the green subpixel SPg on the HTL **822**.
(191) The red HTL **824***r* and the green HTL **824***g* may adjust a length of an optical path in the red, green and blue subpixels SPr, SPg and SPb to obtain a micro cavity.
(192) An electron blocking layer (EBL) **826** is disposed on a whole of the red HTL **824***r* of the red subpixel SPr, the green HTL **824***g* of the green subpixel SPg and the HTL **822** of the blue subpixel SPb.
(193) A first red emitting material layer (EML) **830***r*, a second red emitting material layer (EML) **832***r* and a third red emitting material layer (EML) **834***r* may be sequentially disposed on the EBL **826** of the red subpixel SPr, or a first red emitting material layer (EML) **830***r*, a second red emitting material layer (EML) **832***r* and a third red emitting material layer (EML) **836***r* may be sequentially disposed on the EBL **826** of the red subpixel SPr.
(194) A thickness of the first red EML **830***r* is greater than or equal to a thickness of the second red EML **832***r*, and a thickness of the third red EML **834***r* and **836***r* is smaller than a thickness of the second red EML **832***r*.
(195) A first green emitting material layer (EML) **830***g*, a second green emitting material layer (EML) **832***g* and a third green emitting material layer (EML) **834***g* may be sequentially disposed on the EBL **826** of the green subpixel SPg, or a first green emitting material layer (EML) **830***g*, a second green emitting material layer (EML) **832***g* and a third green emitting material layer (EML) **836***g* may be sequentially disposed on the EBL **826** of the green subpixel SPg.
(196) A thickness of the first green EML **830***g* is greater than or equal to a thickness of the second green EML **832***g*, and a thickness of the third green EML **834***g* and **836***g* is smaller than a thickness of the second green EML **832***g*.
(197) The blue EML **830***b* is selectively disposed on the EBL **826** of the blue subpixel SPb.
(198) A total thickness of the first green EML **830***g*, the second green EML **832***g* and the third green EML **834***g* or a total thickness of the first green EML **830***g*, the second green EML **832***g* and the third green EML **836***g* is greater than a thickness of the blue EML **830***b* and is smaller than a total thickness of the first red EML **830***r*, the second red EML **832***r* and the third red EML **834***r* or a total thickness of the first red EML **830***b*, the second red EML **832***r* and the third red EML **836***r*.
(199) The first red EML **830***r* includes a first red host and a red dopant, and the second red EML **832***r* includes a second red host and a red dopant. The third red EML **834***r* includes a red dopant, and the third red EML **836***r* includes a second red host.
(200) The first green EML **830***g* includes a first green host and a green dopant, and the second green EML **832***g* includes a second green host and a green dopant. The third green EML **834***g* includes a green dopant, and the third green EML **836***g* includes a second green host.

(201) The first red EML **830***r*, the second red EML **832***r* and the third red EML **834***r* may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first red EML **830***r*, the second red EML **832***r* and the third red EML **836***r* may have the same structure as those of the second aspect of FIGS. **4** to **6**.
(202) The first green EML **830***g*, the second green EML **832***g* and the third green EML **834***g* may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first green EML **830***g*, the second green EML **832***g* and the third green EML **836***g* may have the same structure as those of the second aspect of FIGS. **4** to **6**.
(203) An energy level of a triplet excited state T**1** of the first red host of the first red EML **830***r* is lower than an energy level of a triplet excited state T**1** of the second red host of the second red EML **832***r.*
(204) An energy level of a triplet excited state T**1** of the first green host of the first green EML **830***g* is lower than an energy level of a triplet excited state T**1** of the second green host of the second green EML **832***g*.
(205) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third red EML **834***r* has a value between an energy level of a LUMO of the second red EML **832***r* and an energy level of a LUMO of the HBL **840**, and an energy level of a highest occupied molecular orbit (HOMO) of the third red EML **834***r* has a value between an energy level of a HOMO of the second red EML **832***r* and an energy level of a HOMO of the HBL **840**.
(206) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third green EML **834***g* has a value between an energy level of a LUMO of the second green EML **832***g* and an energy level of a LUMO of the HBL **840**, and an energy level of a highest occupied molecular orbit (HOMO) of the third green EML **834***g* has a value between an energy level of a HOMO of the second green EML **832***g* and an energy level of a HOMO of the HBL **840**.
(207) As a result, an electron is promptly transmitted to the first red EML **830***r* and the first green EML **830***g* to improve an emission efficiency.
(208) The second red host of the third red EML **836***r* and the second green host of the third green EML **836***g* may include a bipolar type material.
(209) As a result, an electron is promptly transmitted to the first red EML **830***r* and the first green EML **830***g* to improve an emission efficiency.
(210) Further, since the multiple layered EML is formed in the single chamber, the fabrication process is simplified and deterioration due to contamination during transferring or misalignment of a fine metal mask is prevented.
(211) A hole blocking layer (HBL) **840** is disposed on a whole of the third red EML **834***r* and **836***r* of the red subpixel SPr, the third green EML **834***g* and **836***g* of the green subpixel SPg and the blue EML **830***b* of the blue subpixel SPb, and an electron transporting layer (ETL) **842** is disposed on the HBL **840** in the red, green and blue subpixels SPr, SPg and SPb.
(212) First and second cathodes **844** and **846** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the ETL **842**, and first and second capping layers **850** and **852** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the second cathode **846**.
(213) FIG. **13** is a view showing an organic light emitting diode display device according to a ninth aspect of the present disclosure. A part of the ninth aspect the same as the first and second aspects will be omitted.
(214) In FIG. **13**, an organic light emitting diode (OLED) display device **910** according to a ninth aspect of the present disclosure includes red, green and blue subpixels SPr, SPg and SPb.
(215) An anode (not shown) is disposed in each of the red, green and blue subpixels SPr, SPg and SPb on a substrate (not shown). A hole injecting layer (HIL) **920** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the anode, and a hole transporting layer (HTL) **922** is disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the HIL **920**.

(216) In another aspect, a positive type hole transporting layer (P-HTL) instead of the HIL **920** may be disposed between the anode and the HTL **922**.
(217) A red hole transporting layer (HTL) **924***r* is selectively disposed in the red subpixel SPr on the HTL **922**, and a green hole transporting layer (HTL) **924***g* is selectively disposed in the green subpixel SPg on the HTL **922**.
(218) The red HTL **924***r* and the green HTL **924***g* may adjust a length of an optical path in the red, green and blue subpixels SPr, SPg and SPb to obtain a micro cavity.
(219) An electron blocking layer (EBL) **926** is disposed on a whole of the red HTL **924***r* of the red subpixel SPr, the green HTL **924***g* of the green subpixel SPg and the HTL **922** of the blue subpixel SPb.
(220) A first red emitting material layer (EML) **930***r*, a second red emitting material layer (EML) **932***r* and a third red emitting material layer (EML) **934***r* may be sequentially disposed on the EBL **926** of the red subpixel SPr, or a first red emitting material layer (EML) **930***r*, a second red emitting material layer (EML) **932***r* and a third red emitting material layer (EML) **936***r* may be sequentially disposed on the EBL **926** of the red subpixel SPr.
(221) A thickness of the first red EML **930***r* is greater than or equal to a thickness of the second red EML **932***r*, and a thickness of the third red EML **934***r* and **936***r* is smaller than a thickness of the second red EML **932***r*.
(222) A first green emitting material layer (EML) **930***g*, a second green emitting material layer (EML) **932***g* and a third green emitting material layer (EML) **934***g* may be sequentially disposed on the EBL **926** of the green subpixel SPg, or a first green emitting material layer (EML) **930***g*, a second green emitting material layer (EML) **932***g* and a third green emitting material layer (EML) **936***g* may be sequentially disposed on the EBL **926** of the green subpixel SPg.
(223) A thickness of the first green EML **930***g* is greater than or equal to a thickness of the second green EML **932***g*, and a thickness of the third green EML **934***g* and **936***g* is smaller than a thickness of the second green EML **932***g*.
(224) A first blue emitting material layer (EML) **930***b*, a second blue emitting material layer (EML) **932***b* and a third blue emitting material layer (EML) **934***b* may be sequentially disposed on the EBL **926** of the blue subpixel SPb, or a first blue emitting material layer (EML) **930***b*, a second blue emitting material layer (EML) **932***b* and a third blue emitting material layer (EML) **936***b* may be sequentially disposed on the EBL **926** of the blue subpixel SPb.
(225) A thickness of the first blue EML **930***b* is greater than or equal to a thickness of the second blue EML **932***b*, and a thickness of the third blue EML **934***b* and **936***b* is smaller than a thickness of the second blue EML **932***b*.
(226) A total thickness of the first green EML **930***g*, the second green EML **932***g* and the third green EML **934***g* or a total thickness of the first green EML **930***g*, the second green EML **932***g* and the third green EML **936***g* is greater than a total thickness of the first blue EML **930***b*, the second blue EML **932***b* and the third blue EML **934***b* or a total thickness of the first blue EML **930***b*, the second blue EML **932***b* and the third blue EML **936***b* and is smaller than a total thickness of the first red EML **930***r*, the second red EML **932***r* and the third red EML **934***r* or a total thickness of the first red EML **930***b*, the second red EML **932***r* and the third red EML **936***r*.
(227) The first red EML **930***r* includes a first red host and a red dopant, and the second red EML **932***r* includes a second red host and a red dopant. The third red EML **934***r* includes a red dopant, and the third red EML **936***r* includes a second red host.
(228) The first green EML **930***g* includes a first green host and a green dopant, and the second green EML **932***g* includes a second green host and a green dopant. The third green EML **934***g* includes a green dopant, and the third green EML **936***g* includes a second green host.
(229) The first blue EML **930***b* includes a first blue host and a blue dopant, and the second blue EML **932***b* includes a second blue host and a blue dopant. The third blue EML **934***b* includes a blue dopant, and the third blue EML **936***b* includes a second blue host.

(230) The first red EML **930**_r_, the second red EML **932**_r_ and the third red EML **934**_r_ may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first red EML **930**_r_, the second red EML **932**_r_ and the third red EML **936**_r_ may have the same structure as those of the second aspect of FIGS. **4** to **6**.
(231) The first green EML **930**_g_, the second green EML **932**_g_ and the third green EML **934**_g_ may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first green EML **930**_g_, the second green EML **932**_g_ and the third green EML **936**_g_ may have the same structure as those of the second aspect of FIGS. **4** to **6**.
(232) The first blue EML **930**_b_, the second blue EML **932**_b_ and the third blue EML **934**_b_ may have the same structure as those of the first aspect of FIGS. **1** to **3**, and the first blue EML **930**_b_, the second blue EML **932**_b_ and the third blue EML **936**_b_ may have the same structure as those of the second aspect of FIGS. **4** to **6**.
(233) An energy level of a triplet excited state T**1** of the first red host of the first red EML **930**_r_ is lower than an energy level of a triplet excited state T**1** of the second red host of the second red EML **932**_r_.
(234) An energy level of a triplet excited state T**1** of the first green host of the first green EML **930**_g_ is lower than an energy level of a triplet excited state T**1** of the second green host of the second green EML **932**_g_.
(235) An energy level of a triplet excited state T**1** of the first blue host of the first blue EML **930**_b_ is lower than an energy level of a triplet excited state T**1** of the second blue host of the second blue EML **932**_b_.
(236) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third red EML **934**_r_ has a value between an energy level of a LUMO of the second red EML **932**_r_ and an energy level of a LUMO of the HBL **940**, and an energy level of a highest occupied molecular orbit (HOMO) of the third red EML **934**_r_ has a value between an energy level of a HOMO of the second red EML **932**_r_ and an energy level of a HOMO of the HBL **940**.
(237) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third green EML **934**_g_ has a value between an energy level of a LUMO of the second green EML **932**_g_ and an energy level of a LUMO of the HBL **940**, and an energy level of a highest occupied molecular orbit (HOMO) of the third green EML **934**_g_ has a value between an energy level of a HOMO of the second green EML **932**_g_ and an energy level of a HOMO of the HBL **940**.
(238) An energy level of a lowest unoccupied molecular orbit (LUMO) of the third blue EML **934**_b_ has a value between an energy level of a LUMO of the second blue EML **932**_b_ and an energy level of a LUMO of the HBL **940**, and an energy level of a highest occupied molecular orbit (HOMO) of the third blue EML **934**_b_ has a value between an energy level of a HOMO of the second blue EML **932**_b_ and an energy level of a HOMO of the HBL **940**.
(239) As a result, an electron is promptly transmitted to the first red EML **930**_r_, the first green EML **930**_g_ and the first blue EML **930**_b_ to improve an emission efficiency.
(240) The second red host of the third red EML **936**_r_, the second green host of the third green EML **936**_g_ and the second blue host of the third blue EML **936**_b_ may include a bipolar type material.
(241) As a result, an electron is promptly transmitted to the first red EML **930**_r_, the first green EML **930**_g_ and the first blue EML **930**_b_ to improve an emission efficiency.
(242) Further, since the multiple layered EML is formed in the single chamber, the fabrication process is simplified and deterioration due to contamination during transferring or misalignment of a fine metal mask is prevented.
(243) A hole blocking layer (HBL) **940** is disposed on a whole of the third red EML **934**_r_ and **936**_r_ of the red subpixel SPr, the third green EML **934**_g_ and **936**_g_ of the green subpixel SPg and the third blue EML **934**_b_ and **936**_b_ of the blue subpixel SPb, and an electron transporting layer (ETL) **942** is disposed on the HBL **940** in the red, green and blue subpixels SPr, SPg and SPb.
(244) First and second cathodes **944** and **946** are sequentially disposed in a whole of the red, green

and blue subpixels SPr, SPg and SPb on the ETL **942**, and first and second capping layers **950** and **952** are sequentially disposed in a whole of the red, green and blue subpixels SPr, SPg and SPb on the second cathode **946**.
(245) Consequently, in the OLED display device according to the present disclosure, since the triple layered EML of the first EML of the first host and the dopant, the second EML of the second host and the dopant and the third EML of the dopant is formed in the single chamber, the efficiency is improved and the productivity increases.
(246) Further, since the triple layered EML of the first EML of the first host and the dopant, the second EML of the second host and the dopant and the third EML of the second host is formed in the single chamber, the efficiency is improved and the productivity increases.
(247) It will be apparent to those skilled in the art that various modifications and variation can be made in the present disclosure without departing from the scope of the disclosure. Thus, it is intended that the present disclosure cover the modifications and variations of this disclosure provided they come within the scope of the appended claims.

Claims

1. An organic light emitting diode display device, comprising: a substrate having a plurality of subpixels; and an emitting material layer in the plurality of subpixels disposed on the substrate, wherein the emitting material layer in at least one of the plurality of subpixels comprises: a first emitting material layer including a first host and a dopant; a second emitting material layer on the first emitting material layer and including a second host and the dopant; and a third emitting material layer disposed on the second emitting material layer and including one of the dopant and the second host.
2. The display device of claim 1, wherein a thickness of the third emitting material layer is smaller than a thickness of the second emitting material layer.
3. The display device of claim 1, further comprising a hole blocking layer disposed on the third emitting material layer that includes the dopant, wherein an energy level of a lowest unoccupied molecular orbit of the third emitting material layer has a value between an energy level of a lowest unoccupied molecular orbit of the second emitting material layer and an energy level of a lowest unoccupied molecular orbit of the hole blocking layer, and wherein an energy level of a highest occupied molecular orbit of the third emitting material layer has a value between an energy level of a highest occupied molecular orbit of the second emitting material layer and an energy level of a highest occupied molecular orbit of the hole blocking layer.
4. The display device of claim 1, wherein the third emitting material layer includes the second host, and the second host has a bipolar type.
5. The display device of claim 1, wherein an energy level of a triplet excited state of the first host of the first emitting material layer is smaller than an energy level of a triplet excited state of the second host of the second emitting material layer.
6. The display device of claim 1, wherein the plurality of subpixels includes red, green and blue subpixels, wherein the first host is a first blue host, the dopant is a blue dopant, and the second host is a second blue host, wherein the emitting material layers of the red and green subpixels are red and green emitting material layers, respectively, and wherein the emitting material layer of the blue subpixel includes a first blue emitting material layer having the first blue host and the blue dopant, a second blue emitting material layer having the second blue host and the blue dopant and a third blue emitting material layer having one of the blue dopant and the second blue host.
7. The display device of claim 1, wherein the plurality of subpixels includes red, green and blue subpixels, wherein the first host is a first green host, the dopant is a green dopant, and the second host is a second green host, wherein the emitting material layers of the red and blue subpixels are red and blue emitting material layers, respectively, and wherein the emitting material layer of the

green subpixel includes a first green emitting material layer having the first green host and the green dopant, a second green emitting material layer having the second green host and the green dopant and a third green emitting material layer having one of the green dopant and the second green host.

8. The display device of claim 1, wherein the plurality of subpixels includes red, green and blue subpixels, wherein the first host is a first red host, the dopant is a red dopant, and the second host is a second red host, wherein the emitting material layers of the green and blue subpixels are green and blue emitting material layers, respectively, and wherein the emitting material layer of the red subpixel includes a first red emitting material layer having the first red host and the red dopant, a second red emitting material layer having the second red host and the red dopant and a third red emitting material layer having one of the red dopant and the second red host.

9. The display device of claim 1, wherein the plurality of subpixels includes red, green and blue subpixels, wherein the first host is a first green host and a first blue host, the dopant is a green dopant and a blue dopant, and the second host is a second green host and a second blue host, wherein the emitting material layer of the red subpixel is a red emitting material layer, wherein the emitting material layer of the green subpixel includes a first green emitting material layer having the first green host and the green dopant, a second green emitting material layer having the second green host and the green dopant and a third green emitting material layer having one of the green dopant and the second green host, and wherein the emitting material layer of the blue subpixel includes a first blue emitting material layer having the first blue host and the blue dopant, a second blue emitting material layer having the second blue host and the blue dopant and a third blue emitting material layer having one of the blue dopant and the second blue host.

10. The display device of claim 1, wherein the plurality of subpixels includes red, green and blue subpixels, wherein the first host is a first red host and a first blue host, the dopant is a red dopant and a blue dopant, and the second host is a second red host and a second blue host, wherein the emitting material layer of the green subpixel is a green emitting material layer, wherein the emitting material layer of the red subpixel includes a first red emitting material layer having the first red host and the red dopant, a second red emitting material layer having the second red host and the red dopant and a third red emitting material layer having one of the red dopant and the second red host, and wherein the emitting material layer of the blue subpixel includes a first blue emitting material layer having the first blue host and the blue dopant, a second blue emitting material layer having the second blue host and the blue dopant and a third blue emitting material layer having one of the blue dopant and the second blue host.

11. The display device of claim 1, wherein the plurality of subpixels includes red, green and blue subpixels, wherein the first host is a first red host and a first green host, the dopant is a red dopant and a green dopant, and the second host is a second red host and a second green host, wherein the emitting material layer of the blue subpixel is a blue emitting material layer, wherein the emitting material layer of the red subpixel includes a first red emitting material layer having the first red host and the red dopant, a second red emitting material layer having the second red host and the red dopant and a third red emitting material layer having one of the red dopant and the second red host, and wherein the emitting material layer of the green subpixel includes a first green emitting material layer having the first green host and the green dopant, a second green emitting material layer having the second green host and the green dopant and a third green emitting material layer having one of the green dopant and the second green host.

12. The display device of claim 1, wherein the plurality of subpixels includes red, green and blue subpixels, wherein the first host is a first red host, a first green host and a first blue host, the dopant is a red dopant, a green dopant and a blue dopant, and the second host is a second red host, a second green host and a second blue host, wherein the emitting material layer of the red subpixel includes a first red emitting material layer having the first red host and the red dopant, a second red emitting material layer having the second red host and the red dopant and a third red emitting

material layer having one of the red dopant and the second red host, wherein the emitting material layer of the green subpixel includes a first green emitting material layer having the first green host and the green dopant, a second green emitting material layer having the second green host and the green dopant and a third green emitting material layer having one of the green dopant and the second green host, and wherein the emitting material layer of the blue subpixel includes a first blue emitting material layer having the first blue host and the blue dopant, a second blue emitting material layer having the second blue host and the blue dopant and a third blue emitting material layer having one of the blue dopant and the second blue host.

13. The display device of claim 1, wherein the plurality of subpixels includes red, green and blue subpixels and further includes: an anode in a whole of the red, green and blue subpixels disposed on the substrate; one of a hole injecting layer and a positive type hole transporting layer in a whole of the red, green and blue subpixels disposed on the anode; a hole transporting layer in a whole of the red, green and blue subpixels disposed on one of the hole injecting layer and the positive type hole transporting layer; red and green hole transporting layers in the red and green subpixels, respectively, disposed on one of the hole injecting layer and the positive type hole transporting layer; an electron blocking layer disposed on a whole of the red and green hole transporting layers of the red and green subpixels and the hole transporting layer of the blue subpixel; a hole blocking layer disposed on a whole of the emitting material layer of the red, green and blue subpixels; an electron transporting layer disposed on a whole of the hole blocking layer of the red, green and blue subpixels; first and second cathodes sequentially disposed on a whole of the electron transporting layer of the red, green and blue subpixels; and first and second capping layers sequentially disposed on a whole of the second cathode of the red, green and blue subpixels.

14. The display device of claim 1, wherein the plurality of subpixels includes red, green and blue subpixels, and wherein a thickness of the emitting material layer of the green subpixel is greater than a thickness of the emitting material layer of blue subpixel and is smaller than a thickness of the emitting material layer of the red subpixel.